

# SHAHADAT HOSSAIN

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🌐 Personal Website    in LinkedIn    🐙 GitHub

## EDUCATION

**PhD in Agricultural and Applied Economics** **Exp. Aug 2026**

University of Illinois Urbana-Champaign (UIUC)

Advisors: Alex Winter-Nelson  
Catalina Herrera Almanza

**Master of Social Science in Economics** **Dec 2015**

University of Dhaka, Bangladesh

**Bachelor of Social Science in Economics** **Dec 2014**

University of Dhaka, Bangladesh

## RESEARCH INTERESTS

Agricultural Economics · Applied Econometrics · Causal Inference · Remote Sensing Applications in Agriculture · Geospatial Data Analytics · Agricultural Finance and Insurance · Conservation Agriculture · Agricultural Policy · Food Systems Analysis · Rural Development

## RESEARCH AND ACADEMIC EXPERIENCE

**Graduate Research Assistant** **Jan 2021-Present**

*Department of Agricultural and Consumer Economics, University of Illinois Urbana-Champaign*

- Conduct impact evaluations and causal inference analyses using large-scale household, experimental, administrative, and geospatial datasets across Bangladesh, India, Tanzania, Kenya, and Ethiopia.
- Develop theory-guided econometric and structural models to study agricultural systems, commodity markets, labor allocation, household behavior, and rural development.
- Collaborate with economists, statisticians, public health researchers, survey teams, and other interdisciplinary partners on policy-relevant research and manuscript development.
- Develop reproducible analytical workflows, statistical models, and data-driven decision tools for agricultural and development research.
- Prepare manuscripts for peer-reviewed journals, present findings to academic and policy audiences, and contribute to the development of new research ideas and project proposals.

**Short-term Consultant** **Jul 2024-Dec 2024**

*Development Economics Research Group, World Bank*

Project: Women's Mental Health and Household Dynamics (India)

- Conducted impact evaluation and applied econometric analyses using household survey, administrative, experimental, and geospatial datasets to support evidence-based development policy.
- Cleaned, harmonized, integrated, and analyzed large-scale datasets using Stata and R and prepared technical reports, visualizations, and presentations for multidisciplinary research teams.
- Contributed to a research project accepted for publication in the *Journal of Development Economics*.

**Research Consultant**

*Bangladesh Institute of Development Studies*

### Selected Projects:

Post-Enumeration Check of Bangladesh Economic Census 2025 **Jul 2025-Dec 2025**  
Post-Enumeration Check of Bangladesh Population and Housing Census 2022 **Nov 2023-Jul 2024**

Rural Livelihood Dynamics in the Haor Region of Bangladesh **Sep 2022-Nov 2022**

- Cleaned, validated, and analyzed the Post Enumeration Survey dataset of the Housing and Population Census 2022 and the Economic Census 2025 in Bangladesh using the enumeration area shape files.
- Contributed to questionnaire design, household and community survey implementation, fieldwork coordination, real-time monitoring, and data quality assurance.
- Co-authored the Post Enumeration Survey Report.

### Research Analyst

**Nov 2017-Jan 2021**

*Development Strategy and Governance Unit, International Food Policy Research Institute (IFPRI)*

Project: “Bangladesh Modern Food Storage Facilities Project”

- Developed stochastic spatial equilibrium and econometric models to evaluate agricultural price stabilization, grain storage, market risk, trade, food security, and agricultural policy alternatives.
- Analyzed commodity prices, production, inventories, and market integration using nationally representative survey and administrative datasets.
- Developed structural supply, demand, market equilibrium, and production models to estimate equilibrium prices, producer responses, farm profitability, and the impacts of agricultural policy and commodity price changes.
- Produced policy reports and technical publications used by government agencies and international organizations.
- Coordinated field surveys, managed research workflows, collaborated with multidisciplinary teams, and contributed to grant-supported research projects and technical deliverables.

### Research Associate

**Mar 2017-Oct 2017**

*BRAC Institute of Governance and Development (BIGD)*

Project: “State of Governance Report 2017”

- Assisted in questionnaire design, household survey implementation, and econometric analysis.
- Performed data cleaning and econometric analysis for the report.
- Contributed to report writing.

## SELECTED RESEARCH

**S. Hossain.** “Commodity Price Dynamics, Shadow Wages, and Rural Labor Supply: Evidence from Bangladesh” [*Job Market Paper*] [Link](#) 

This paper develops a non-separable agricultural household model to examine how poultry price shocks affect shadow wages, household income, and labor allocation. Using three waves of nationally representative panel data and instrumental variables, it estimates gender-specific labor responses to commodity price changes and highlights the role of imperfect labor markets in shaping rural household behavior.

S. Anukriti, C. Herrera-Almanza, **S. Hossain**, and M. Karra. “Son Preference, Intergenerational Household Dynamics, and Women’s Mental Health in India” [*Journal of Development Economics*] [Link](#) 

Examines how son preference affects women’s mental health and intergenerational household dynamics in India using quasi-experimental methods. The study shows that having a first-born son improves mothers-in-law’s mental health, increases childcare provided by mothers-in-law, and raises daughters-in-law’s labor force participation.

**S. Hossain.** “Impacts of illness and job loss on households within economy-wide shocks: Coping through labor markets”

Uses a two-wave household panel and difference-in-differences with matching to estimate how COVID-19-related illness and job loss affected income, food security, assets, labor supply, and household coping strategies in rural Bangladesh.

“Crop Rotation and Crop Diversity Analysis Using Satellite Imagery in California Agriculture”, Graduate Big Data Analytics Project.

In the project, I used Landsat satellite imagery and geospatial data analytics to examine crop rotation patterns and measure crop diversity across agricultural areas in California using Python and R. In addition, I processed and analyzed spatially referenced agricultural data and developed visualizations to characterize land-use and cropping patterns.

## RESEARCH PUBLICATIONS

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Minot, Nicholas; **Hossain, Shahadat**; Kabir, Razin; Dorosh, Paul A.; and Rashid, Shahidur. 2024. “Assessing the impact of rice price stabilization policies in Bangladesh: Results from a stochastic spatial equilibrium model”. IFPRI Discussion Paper 2252. Washington, DC: International Food Policy Research Institute. [Link](#)

Minot, Nicholas; **Hossain, Shahadat**; Kabir, Razin; Dorosh, Paul A.; and Rashid, Shahidur. 2021. “Rice price stabilization in Bangladesh: Assessing the impact of public farm-gate and consumer price stabilization instruments on the overall grain market and developing orientations with a greater role for the private sector”. IFPRP Working Paper 11. Washington, DC: International Food Policy Research Institute (IFPRI). [Link](#)

Dorosh, Paul A.; Minot, Nicholas; Kabir, Razin; and **Hossain, Shahadat**. 2021. “Private sector rice stocks in Bangladesh: Estimates from the Bangladesh Millers’ and Traders’ Survey (MATS) 2018”. IFPRP Working Paper 6. Washington, DC: International Food Policy Research Institute (IFPRI). [Link](#)

Chowdhury, Reajul Alam; Alam, Md Monjurul; Ali, Md Rostom; Awal, Md Abdul; **Hossain, Shahadat**; Kalita, Prasanta Kumar; Saha, Chayan Kumer; Winter-Nelson, Alex. 2023. “Private investments in modern foodgrain storage in Bangladesh: An economic feasibility analysis and potential role for the public sector”. IFPRP Working Paper 12. Washington, DC: International Food Policy Research Institute (IFPRI). [Link](#)

## OUTREACH ARTICLES

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**S. Hossain** and J. Janzen. “The Impact of Long-Run Declines in Gasoline Use on the US Corn Market.” *farmdoc daily* (12):195, Department of Agricultural and Consumer Economics, University of Illinois at Urbana-Champaign, December 27, 2022. [Link](#)

## TEACHING EXPERIENCE

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### Teaching Assistant - University of Illinois Urbana-Champaign

Courses Assisted:

- ACE 100: Introduction to Applied Microeconomics (Fall 2024)
  - Led interactive discussion sessions with a class of 35 students
- ACE 440: Financial Plan for Professionals (Spring 2025)
- ACE 251: World Food Economy (Fall 2025)
  - Led three weekly interactive discussion sessions of 35 students each.

- Assessed group projects through detailed review, feedback, and grading.
- ACE 231: Food and Agribusiness Management (Spring 2026)
- Assessed group projects through detailed review, feedback, and grading.

**Instructor - BRAC Institute of Governance and Development, BRAC University**

Course Taught:

- Applied Statistical Methods (Summer 2017)

## CONFERENCE PRESENTATIONS

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### Poster presentation

“Impacts of COVID-19 illness and job loss on household welfare: coping through labor markets” at the Agricultural and Applied Economics Association (AAEA), Denver, CO **Jul 2025**

### Oral presentation

“Impacts of COVID-19 illness and job loss on household welfare: coping through labor markets” at International Policy and Development (IPAD), Agriculture and Consumer Economics, UIUC **Mar 2024**

“Commodity Price Dynamics, Shadow Wages, and Rural Labor Supply: Evidence from Bangladesh” at International Policy and Development (IPAD), Agriculture and Consumer Economics, UIUC **Oct 2025**

## FELLOWSHIPS AND AWARDS

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**AAEA Travel Grant** - International Section, Agricultural and Applied Economics Association (AAEA) **Summer 2025**

**Louise Scholars Graduate Student Fellowship** - Agriculture and Consumer Economics, University of Illinois Urbana-Champaign **Fall 2024**

**Post Harvest Loss Prevention Fellow** - ADM Institute for the Prevention of Postharvest Loss, University of Illinois Urbana-Champaign **Spring 2021 - Summer 2023**

## PROFESSIONAL SERVICE

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**Peer Reviewer:** *Studies in Family Planning*; Midwest International Economic Development Conference

## TECHNICAL SKILLS

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**Programming & Geospatial Tools:** R, Python, ArcGIS, Stata, SQL, Git, Microsoft Excel, SurveyCTO

**Econometrics & Causal Inference:** Difference-in-Differences, Instrumental Variables, Fixed Effects, Panel Data, Matching, Regression Discontinuity, Randomized Controlled Trials, Structural Modeling, and Machine-Learning-Assisted Estimation

**Geospatial Analysis & Remote Sensing:** Satellite imagery analysis, agricultural remote sensing, spatial data processing, geospatial datasets, land-use and crop-pattern analysis, spatial visualization

**Data Management:** Household surveys, market surveys, administrative data, panel data, experimental data, geospatial data, and reproducible workflows

**Visualization:** R Shiny, Tableau, ggplot2

**Languages:** English (Fluent), Bangla (Native), Hindi (Intermediate)